Earthquake-Resistant Building Design Concepts

Cover

Mini Project Report B.Tech Civil Engineering

Abstract

This report explains key concepts of earthquake-resistant building design including structural planning, load paths, ductility, and seismic detailing.

Introduction

Earthquakes induce lateral forces that can severely damage poorly designed structures. Seismic design aims to protect life and reduce damage.

Objectives

1. Understand seismic loads.
2. Explain earthquake-resistant principles.
3. Discuss materials and detailing.
4. Recommend safe design practices.

Literature Review

Modern seismic codes emphasize ductility, regular building configuration, shear walls, braced frames, and base isolation for important structures.

Methodology

Study IS 1893 and IS 13920 concepts, compare structural systems, and summarize engineering recommendations.

Design/Analysis

Key concepts: symmetric floor plans, strong column–weak beam philosophy, shear walls, confined reinforcement, lightweight materials, proper foundations, and quality construction. Simple load path diagram:

Roof → Beams → Columns/Shear Walls → Foundation → Soil

Diagrams/Charts

Flow Chart:
Earthquake → Ground Motion → Structural Response → Seismic Forces → Safe Design Measures

Conclusion

Earthquake-resistant design improves safety through proper planning, ductile detailing, and code-compliant structural systems.

References

IS 1893 (Part 1), IS 13920, NBC India, standard structural engineering textbooks.